

Computer Vision Practical Report:

VIDEO IN REVERSE MODE

1. AIM

To read an input video file using OpenCV, store all individual frames into a sequential array/list, and render/save the video frames in reverse chronological order.

2. PROCEDURE

- Open the source video file using `cv2.VideoCapture()`.
- Read frame-by-frame inside a while loop using `cap.read()` and append each valid frame into a Python list.
- Release the capture object (`cap.release()`).
- Iterate backwards through the frame list (or reverse the list using `frames.reverse()`).
- Display each frame sequentially using `cv2.imshow()` with appropriate delay via `cv2.waitKey()`, or write the reversed sequence to an output file using `cv2.VideoWriter()`.

3. PROGRAM CODE

```
import cv2

# 1. Open the video file
video_path = 'E37-VID.mp4'
cap = cv2.VideoCapture(video_path)

if not cap.isOpened():
    print(f"Error: Could not open video '{E37-VID}'")
else:
    # 2. Read all frames into a list
    frames = []
    while True:
        ret, frame = cap.read()
        if not ret:
```

```
        break

    frames.append(frame)

cap.release()
print(f"Total frames loaded: {len(frames)}")

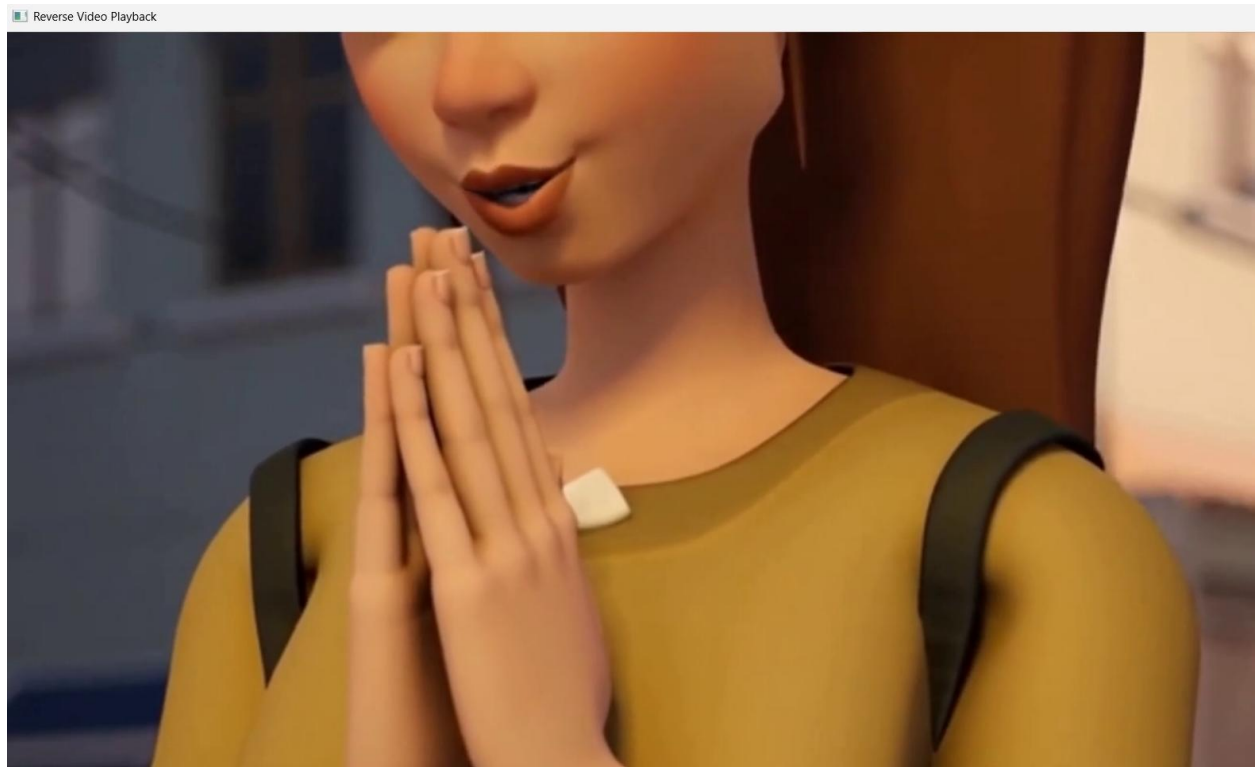
# 3. Get original video FPS (Frames Per Second) to maintain correct playback speed
fps = int(cap.get(cv2.CAP_PROP_FPS))
if fps == 0:
    fps = 30 # Default fallback if FPS read fails
delay = int(1000 / fps)

# 4. Play frames in reverse order
for frame in reversed(frames):
    cv2.imshow('Reverse Video Playback', frame)

    # Press 'q' on keyboard to exit early
    if cv2.waitKey(delay) & 0xFF == ord('q'):
        break

cv2.destroyAllWindows()
```

4. OUTPUT



5. RESULT

The Python script executed successfully. The input image was loaded.